```cpp
#include "FibonacciSequence.h"

FibonacciSequence::FibonacciSequence() noexcept
    : fPrevious(0), fCurrent(1){}

const uint64_t& FibonacciSequence::operator*() const noexcept {
    return fCurrent;
}

FibonacciSequence& FibonacciSequence::operator++() noexcept {
    fCurrent = fCurrent + fPrevious;
    fPrevious = fCurrent - fPrevious;
    return *this; //object itself
}

// what is this?
FibonacciSequence FibonacciSequence::operator++(int) noexcept {
    FibonacciSequence temp = *this;
    ++(*this);
    return temp;
}

bool FibonacciSequence::operator==(const FibonacciSequence& aOther) const noexcept {
    return fPrevious == aOther.fPrevious && fCurrent == aOther.fCurrent;
}

void FibonacciSequence::begin() noexcept {
    fPrevious = 0;
    fCurrent = 1;
}

void FibonacciSequence::end() noexcept {
    fPrevious = 0;
    fCurrent = 0;
}
```